RAG-based QA System using Wikipedia pages (Cricket Domain)

Name:Vaishnav Nigade Roll no:2022BCD0045

Installing required libraries

```
!pip install -q langchain langchain-community langchain-huggingface chromadb wikipedia-api senter

# Optional: if you want cleaner output in notebooks
!pip install -q ipywidgets
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 42.2/42.2 kB 1.1 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 52.0/52.0 kB 3.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.5/2.5 MB 20.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 21.6/21.6 MB 52.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 76.0/76.0 kB 7.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.7/60.7 MB 13.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 278.2/278.2 kB 21.5 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.0/2.0 MB 75.7 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 47.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 17.2/17.2 MB 62.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 72.1/72.1 kB 5.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 142.0/142.0 kB 8.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 68.7/68.7 kB 6.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 231.6/231.6 kB 18.7 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71.6/71.6 kB 5.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.6/60.6 kB 5.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 65.0/65.0 kB 3.6 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 51.0/51.0 kB 4.5 MB/s eta 0:00:00
ERROR: pip's dependency resolver does not currently take into account all the packages that are in
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.33.0 which is incompatible.
google-adk 1.27.1 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.
google-adk 1.27.1 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you ha
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have c
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have ope
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.6/1.6 MB 11.5 MB/s eta 0:00:00
```

2. Imports

```
import os
import wikipediaapi

from langchain_core.documents import Document
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.vectorstores import Chroma
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_core.prompts import PromptTemplate
from langchain_core.runnables import RunnablePassthrough
from langchain_core.output_parsers import StrOutputParser

from transformers import pipeline
from langchain_huggingface import HuggingFacePipeline
```

3. Choosing your domain pages

```
# Domain: Cricket
wiki_titles = [
    "Cricket",
```

```
    "Virat Kohli",
    "Sachin Tendulkar",
    "MS Dhoni",
    "Rohit Sharma",
    "Indian Premier League",
    "Cricket World Cup",
    "Test cricket",
    "One Day International",
    "Twenty20 International"
]
```

4. Crawl data from Wikipedia

```
wiki = wikipediaapi.Wikipedia(
    language="en",
    user_agent="rag-cricket-qa-system/1.0"
)

documents = []

for title in wiki_titles:
    page = wiki.page(title)
    if page.exists():
        content = page.text.strip()
        if content:
            documents.append(
                Document(
                    page_content=content,
                    metadata={"source": f"Wikipedia - {title}", "title": title}
                )
            )
            print(f"Loaded: {title}")
    else:
        print(f"Page not found: {title}")

print(f"\nTotal documents loaded: {len(documents)}")
```

```
Loaded: Cricket
Loaded: Virat Kohli
Loaded: Sachin Tendulkar
Loaded: MS Dhoni
Loaded: Rohit Sharma
Loaded: Indian Premier League
Loaded: Cricket World Cup
Loaded: Test cricket
Loaded: One Day International
Loaded: Twenty20 International

Total documents loaded: 10
```

5. Split documents into chunks

```
text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=1000,
    chunk_overlap=200
)

split_docs = text_splitter.split_documents(documents)

print(f"Total chunks created: {len(split_docs)}")
print("\nSample chunk:\n")
print(split_docs[0].page_content[:800])
```

```
Total chunks created: 557
```

```
Sample chunk:

Cricket is a bat-and-ball game that is played between two teams of eleven players on a field, at t
```

6. Create embeddings

```
embedding_model = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingfac
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datas
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 22.0kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable h
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hu
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 10.1kB/s]
README.md: 10.5k/? [00:00<00:00, 512kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 5.17kB/s]
config.json: 100%  612/612 [00:00<00:00, 49.3kB/s]
model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 458MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 431.94it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expec
tokenizer_config.json: 100%  350/350 [00:00<00:00, 23.1kB/s]
vocab.txt: 232k/? [00:00<00:00, 8.39MB/s]
tokenizer.json: 466k/? [00:00<00:00, 20.0MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 10.2kB/s]
config.json: 100%  190/190 [00:00<00:00, 13.8kB/s]
```

7. Store embeddings in Chroma DB

```
vectorstore = Chroma.from_documents(
    documents=split_docs,
    embedding=embedding_model,
    persist_directory="./chroma_cricket_db"
)

retriever = vectorstore.as_retriever(search_kwargs={"k": 4})
```

8. Load a text generation model

```
from transformers import pipeline
from langchain_huggingface import HuggingFacePipeline

generator = pipeline(
```

```
    "text-generation",
    model="google/flan-t5-base",
    max_new_tokens=256,
    temperature=0.2
)

llm = HuggingFacePipeline(pipeline=generator)
```

```
model.safetensors: 100%                                    990M/990M [00:16<00:00, 65.6MB/s]

Loading weights: 100%                    282/282 [00:00<00:00, 611.33it/s, Materializing param=shared.weight]
The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weigh
generation_config.json: 100%                                    147/147 [00:00<00:00, 6.31kB/s]

tokenizer_config.json:      2.54k/? [00:00<00:00, 109kB/s]

spiece.model: 100%                                    792k/792k [00:00<00:00, 3.95MB/s]

tokenizer.json:      2.42M/? [00:00<00:00, 2.44MB/s]

special_tokens_map.json:      2.20k/? [00:00<00:00, 99.1kB/s]
Passing `generation_config` together with generation-related arguments=({'max_new_tokens', 'temper
The model 'T5ForConditionalGeneration' is not supported for text-generation. Supported models are
```

9. Define RAG prompt

```
prompt_template = """
You are a helpful question-answering assistant.

Answer the question only from the provided context.
If the answer is not present in the context, say:
"I could not find the answer in the retrieved documents."

Context:
{context}

Question:
{question}

Answer:
"""

prompt = PromptTemplate(
    template=prompt_template,
    input_variables=["context", "question"]
)
```

10. Helper function to format docs

```
def format_docs(docs):
    return "\n\n".join(
        [f"Source: {doc.metadata.get('title', 'Unknown')}\n{doc.page_content}" for doc in docs]
    )
```

11. Build the RAG chain

```
rag_chain = (
    {
        "context": retriever | format_docs,
        "question": RunnablePassthrough()
    }
    | prompt
```

```
        | llm
        | StrOutputParser()
    )
```

12. Ask questions

```
questions = [
    "Who is known as the captain cool in Indian cricket?",
    "What is the role of Virat Kohli in cricket?",
    "What is the Indian Premier League?",
    "How is Test cricket different from T20 cricket?",
    "Who has scored 100 international centuries?"
]

for q in questions:
    print(f"\nQuestion: {q}")
    print("Answer:", rag_chain.invoke(q))
    print("-" * 80)
```

```
Token indices sequence length is longer than the specified maximum sequence length for this mode

Question: Who is known as the captain cool in Indian cricket?
Both `max_new_tokens` (=256) and `max_length`(=20) seem to have been set. `max_new_tokens` will
Answer:
You are a helpful question-answering assistant.

Answer the question only from the provided context.
If the answer is not present in the context, say:
"I could not find the answer in the retrieved documents."

Context:
Source: MS Dhoni
As a wicket-keeper, he has been praised for his fast reflexes behind the stumps while also being

Source: Rohit Sharma
Rohit Gurunath Sharma (born 30 April 1987) is an Indian international cricketer and the former c

Source: MS Dhoni
Individual
MTV Youth Icon of the Year: 2006
ICC Men's ODI Team of the Year: 2006, 2008, 2009, 2010, 2011, 2012, 2013, 2014 (captain in 2009,
ICC Men's Test Team of the Year: 2009, 2010, 2013 (captain in 2009, 2010)
Major Dhyanchand Khel Ratna award: 2008
ICC ODI Cricketer of the Year: 2008, 2009
Padma Shri: 2009
Honorary Lieutenant Colonel, Parachute Regiment of the Indian Territorial Army: 2011
CNN-News18 Indian of the Year: 2011
Castrol Indian Cricketer of the Year: 2011
ICC People's Choice Award: 2013
Padma Bhushan: 2018
ICC Men's ODI team of the decade: 2011–2020 (captain and wicket-keeper)
ICC Men's T20I team of the decade: 2011–2020 (captain and wicket-keeper)
ICC Spirit of cricket award of the decade: 2011–2020
ICC Cricket Hall of Fame: 2025
Others

Source: MS Dhoni
Mahendra Singh  Dhoni ([məˈɦeːnd̪rə ˈsɪŋgʰ ˈd̪ʰoːniː] ; born 7 July 1981) is an Indian professiona

Question:
Who is known as the captain cool in Indian cricket?

Answer:

--------------------------------------------------------------------------------

Question: What is the role of Virat Kohli in cricket?
Both `max_new_tokens` (=256) and `max_length`(=20) seem to have been set. `max_new_tokens` will
Answer:
You are a helpful question-answering assistant.
```

```
Answer the question only from the provided context.
If the answer is not present in the context, say:
"I could not find the answer in the retrieved documents."

Context:
```

Double-click (or enter) to edit

```
query = "Who has scored 100 international centuries?"
retrieved_docs = retriever.invoke(query)

for i, doc in enumerate(retrieved_docs, 1):
    print(f"\n--- Retrieved Chunk {i} ---")
    print("Source:", doc.metadata.get("title"))
    print(doc.page_content[:700])
```

```
--- Retrieved Chunk 1 ---
Source: Sachin Tendulkar
Centuries
Tendulkar holds the record of the highest number of centuries in Tests (51) and stands second in t

--- Retrieved Chunk 2 ---
Source: Sachin Tendulkar
100th international century
On 16 March 2012, Tendulkar accomplished a remarkable feat by scoring his 100th international cent

--- Retrieved Chunk 3 ---
Source: Sachin Tendulkar
In the fourth Test match of the 2008–09 Border–Gavaskar Trophy against Australia on 6 November 200

--- Retrieved Chunk 4 ---
Source: Virat Kohli
Career summary
Kohli has made 85 centuries which includes 7 double centuries in international cricket—30 centurie

Test match performance
ODI match performance
T20I match performance
Honours
National honours
2013 – Arjuna Award, second highest sporting honour.
2017 –   Padma Shri, India's fourth highest civilian award.
2018 – Rajiv Gandhi Khel Ratna Award, India's highest sporting honour.
```

```
while True:
    user_q = input("\nAsk a cricket question (or type 'exit'): ")
    if user_q.lower() == "exit":
        break
    answer = rag_chain.invoke(user_q)
    print("\nAnswer:", answer)
```